

Linux Process Monitoring and Control System

Project Abstract

1. Problem Statement

Managing and understanding running processes is a fundamental Operating Systems task. Linux provides process information and control mechanisms, but users generally interact with them through separate command-line utilities. This project aims to develop a unified Linux-based system for monitoring running processes and performing basic process-control operations while demonstrating core Operating Systems and Linux/POSIX concepts.

2. Objectives

- Monitor and display information about active Linux processes.
- Observe process states, process IDs, parent-child relationships, and resource usage.
- Provide controlled operations for managing processes.
- Demonstrate Linux process management and communication mechanisms.
- Develop practical understanding of Linux/POSIX system programming.

3. Proposed Approach / Methodology

The system will identify and monitor active processes using Linux interfaces and system-level mechanisms. Process information will be collected and presented in a structured manner. The system will support process-control operations such as sending appropriate signals to selected processes and observing their resulting state changes. Parent-child process relationships and relevant resource information will also be monitored. The implementation will primarily focus on Linux/POSIX mechanisms rather than generic application-development features.

4. Linux/POSIX Technologies and System Calls

- Linux /proc filesystem for process information
- Process IDs (PIDs) and process states
- POSIX/Linux signals
- kill() for sending signals to processes
- fork() for process creation concepts
- exec() for process execution concepts
- wait() for process termination and synchronization concepts
- Standard Linux system interfaces and APIs for process/resource monitoring

5. Expected Outcome

The expected outcome is a working Linux-based Process Monitoring and Control System capable of displaying process information, monitoring process states and resource usage, identifying process relationships, and performing basic process-control operations. The project will demonstrate practical understanding of Linux processes, system calls, signals, process management, and other Operating Systems concepts.